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South East Technological University

THE EVOLUTION OF MACHINE LEARNING AND AI · WEEK 5 OF 12

Teaching a Machine a Law of Nature

Tell it the rule, and it needs no examples

Live lecture · 2 hours · one picture per idea · everything in a red box is quiz material

FOUR THINGS BEFORE WE START

How today works

★ Red boxes are the quiz

Every part ends with one. Take a photo.

One rule, never solved by you

I say it in words. The computer does the rest.

The lab follows the same six parts

Some cells take a minute; the page says which.

Stuck 40 minutes? Post in the channel

That is a bug in the material, not in you.

Today

Tell a machine the rule, and it needs no examples.

Tell it the wrong rule, and it is wrong without warning.

Week 4: a good guess built into a machine is worth a mountain of examples. This week the guess is not "usually true". It is a law of nature: always true.

THE ONE PICTURE TO REMEMBER

Marking homework without the answer sheet

Question: ? × ? = 9

Student wrote: 4



Check: 4 × 4 = 16, not 9

EVERYDAY ANALOGY

You never needed to know the answer was 3. You only needed to **check**. A law of nature is a question you can put an answer back into.

SIX SHORT PARTS · NO BREAK

Today's route

Part 1	A law is not a pattern	⏱ 15 min
Part 2	Checking the rule where there is no data	⏱ 23 min
Part 3	What the rule buys	⏱ 17 min
Part 4	What it costs, and how it goes quietly wrong	⏱ 27 min
Part 5	Four more ways to build in science	⏱ 19 min
Part 6	Choosing	⏱ 9 min

> **KEY IDEA**
> Three times today we read a few lines of real code together. Nothing is run in class; you run it in the lab.

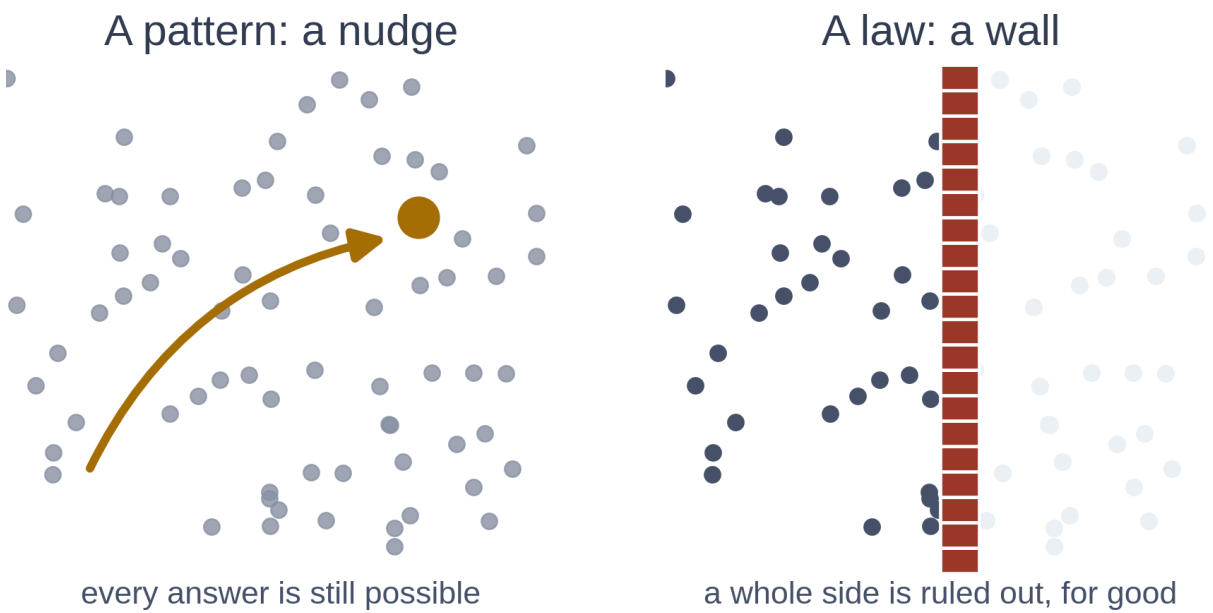
PART 1

A law is not a pattern

Why a rule beats a pile of examples

TWO KINDS OF ADVICE

A pattern nudges. A law rules things out.

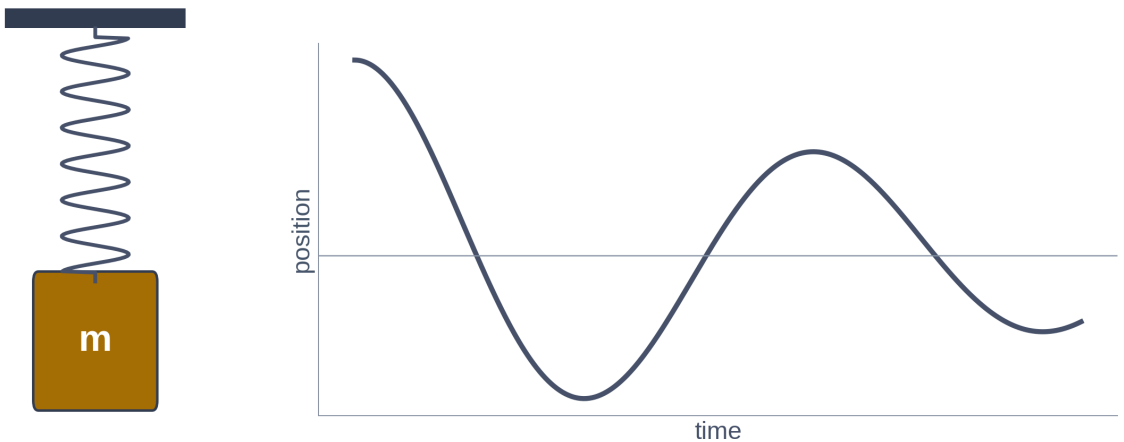


KEY IDEA

Last week, machines got patterns: usually right, ruling nothing out. This week they get laws: a wall, and the wrong side is gone for good.

THE ONLY EQUATION, IN WORDS FIRST

One rule all day: a weight on a spring

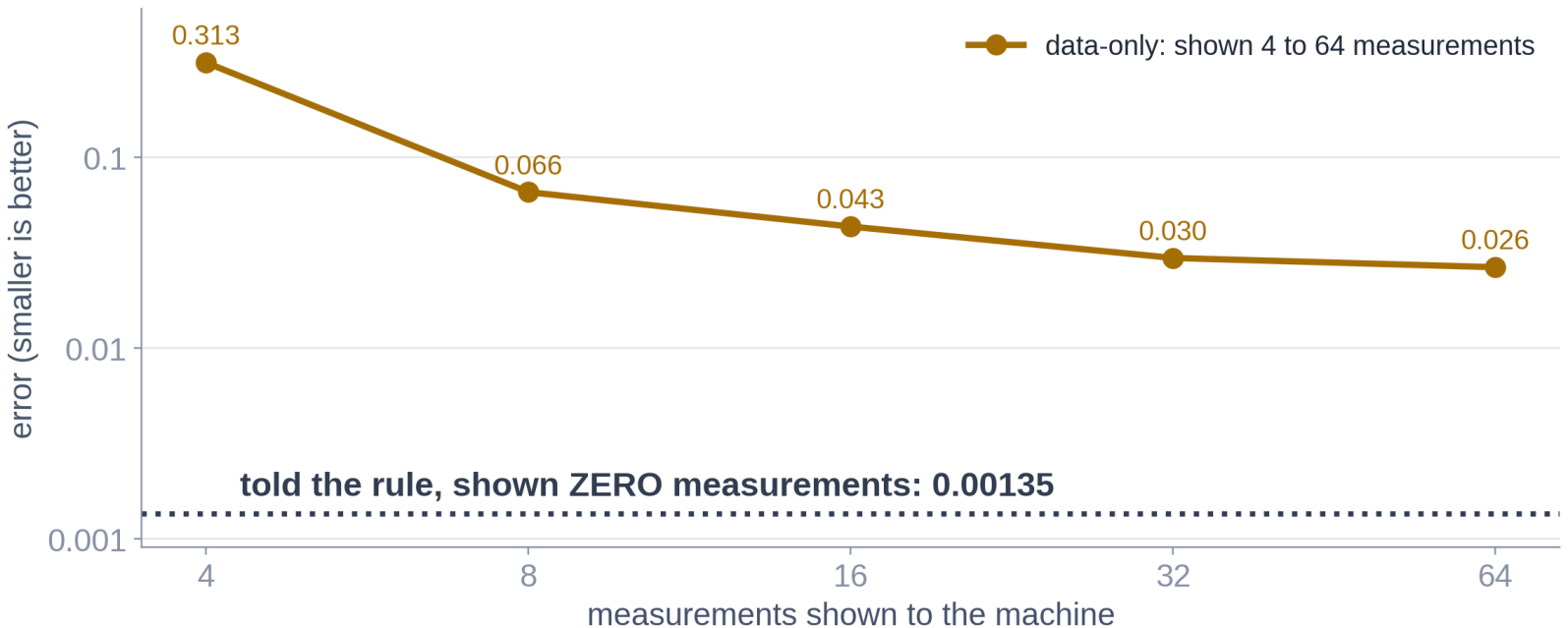


heaviness × change of speed + friction × speed + stiffness × position = 0

In books: $m\,u'' + c\,u' + k\,u = 0$. You will never solve it.

SAME MACHINE · SAME TRAINING · A DIFFERENT CHECK

Told the rule, shown nothing, it wins



KEY IDEA

Shown 64 measurements: error **0.026**. Told the rule and shown **none**: **0.00135**. About 20 times better, from no data at all.

WHY THIS MATTERS TO YOU

Where you will work, examples are scarce

40
readings from one wind turbine

200
patients from one hospital

1
bridge, with a few sensors

KEY IDEA

More data is not for sale. What you already know is free: hundreds of years of reliable rules about springs, heat and flow.

Part 1, done

★ REMEMBER FOR THE WEEK 6 QUIZ

- A pattern nudges. A law rules wrong answers out, for good.
- A law is a question you can put an answer back into: no answer sheet needed.
- Told the rule and shown 0 measurements: error 0.00135. Shown 64: 0.026.
- Where data is scarce, what you already know is the strongest help you have.

PART 2

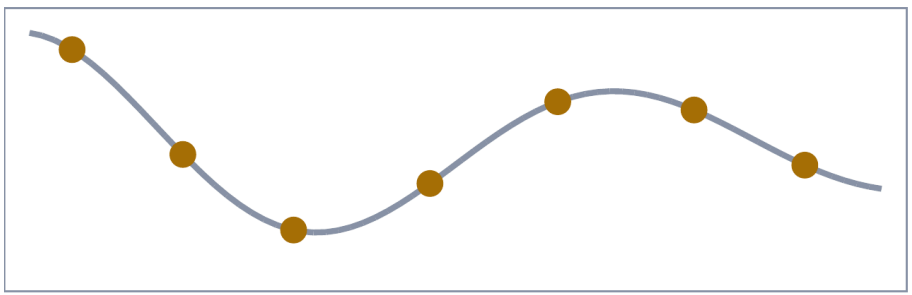
Checking the rule where there is no data

How a machine is checked against a rule, step by step

TWO WAYS TO CHECK A GUESS

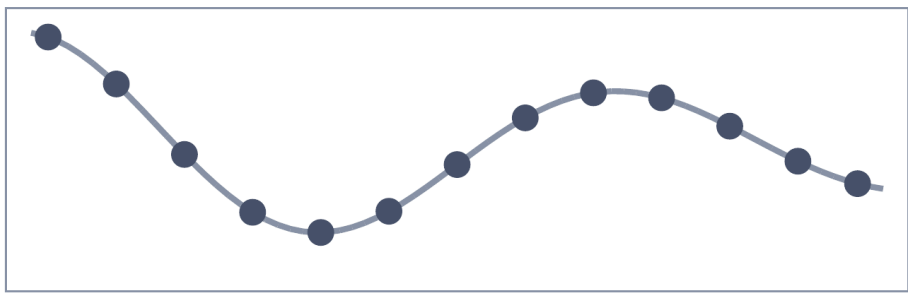
Checks against the rule are free

Data loss



compare against measurements
somebody had to go and collect

Physics loss

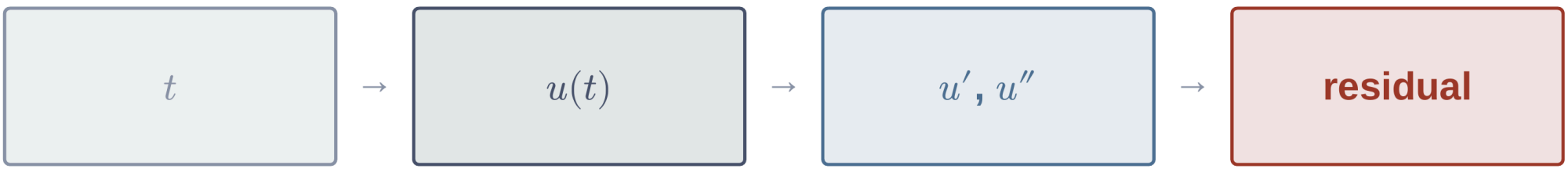


check the equation where there
is nothing to measure at all

Checking the rule does not need the answer. So the checks cost nothing, and they can go anywhere: inside an engine, at the centre of the sun, ten years from now.

FOLLOW THE ARROWS

What one check does



a time, with no measurement attached

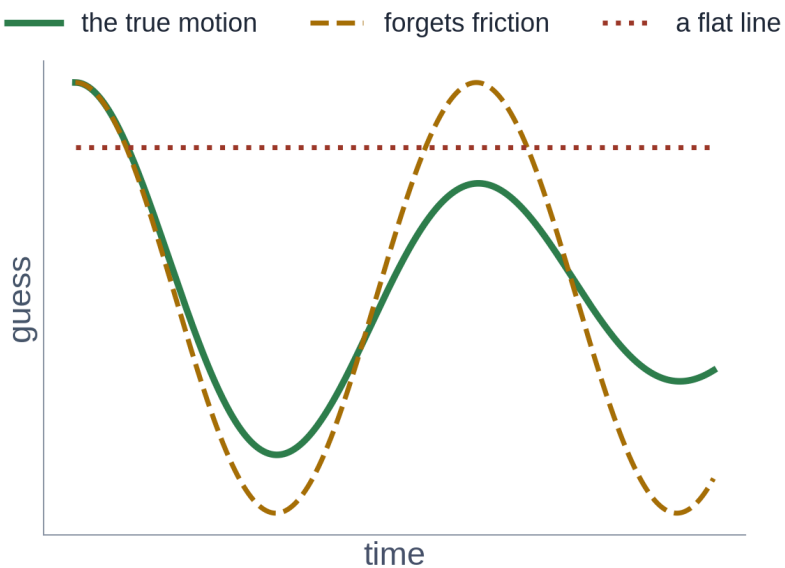
the network's guess

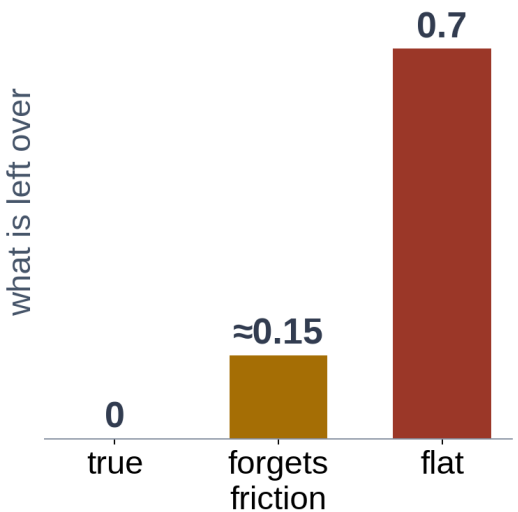
derivatives of the network with respect to t

$mu'' + cu' + ku$, squared and averaged

The true answer appears nowhere. Square the leftover, average it over a few hundred moments: that is the whole score. Then blame and nudge, as in Week 3.

The rule marks the homework





KEY IDEA

True motion: nothing left over. Forgets friction: a little left over. Flat line: a lot. Nobody looked up the answer.

WEEK 3'S LOOP, ROW BY ROW

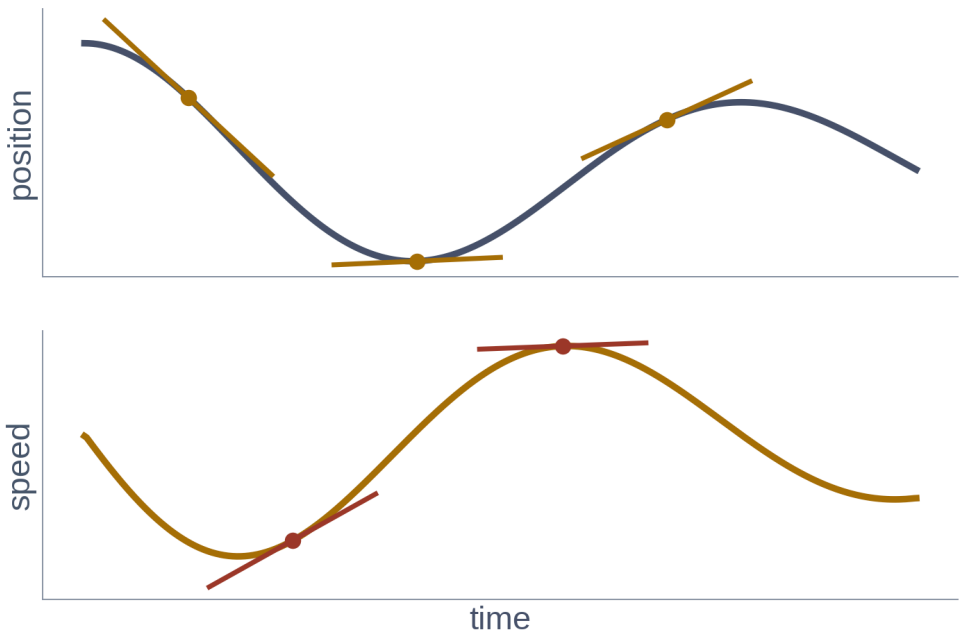
Only one step changed

the loop's step	Weeks 3 and 4	this week
guess	input in, answer out	the same
check	against examples	**against the rule**
blame	which dials caused it	the same
nudge	turn the dials a little	the same

KEY IDEA

A "physics-informed network" is an ordinary network with a different check. There is no special architecture.

The one new trick: the slope of a slope



KEY IDEA

The computer must work out slopes of slopes **exactly**. The lab's small engine does, to rounding error. Nudging and measuring is not good enough.

The code: the check in five lines

week5-lab/pinn.py — python

```python
def rule_check(net, times):
    u   = net(times)              # 1 the guess
    v   = grad(u, times)          # 2 its speed
    a   = grad(v, times)          # 3 the slope of the speed
    left_over = m*a + c*v + k*u   # 4 put all three into the rule
    return mean(left_over ** 2)   # 5 the score: no answer sheet
```

KEY IDEA

`grad` twice is the slope of a slope. Read the five lines again: the true answer is not there.

Part 2, done

★ REMEMBER FOR THE WEEK 6 QUIZ

- The leftover: put the guess back into the rule; zero means it obeys.
- The score is the leftover, squared and averaged at free check points. No answer appears.
- Only the check step changed. A physics-informed network is an ordinary network.
- The one new trick: the computer must take the slope of a slope, exactly.

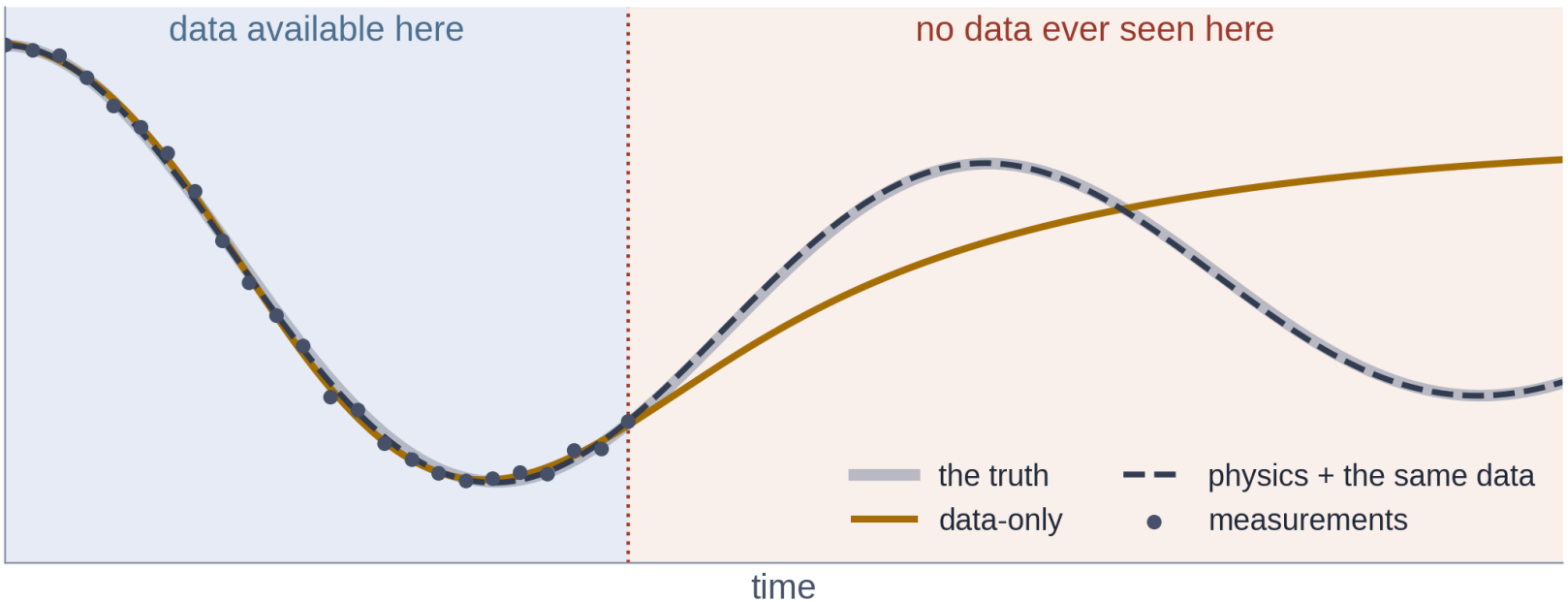
PART 3

What the rule buys

Beyond the data, and something nobody asked for

MEASUREMENTS ON THE LEFT ONLY

Beyond the data, it keeps going

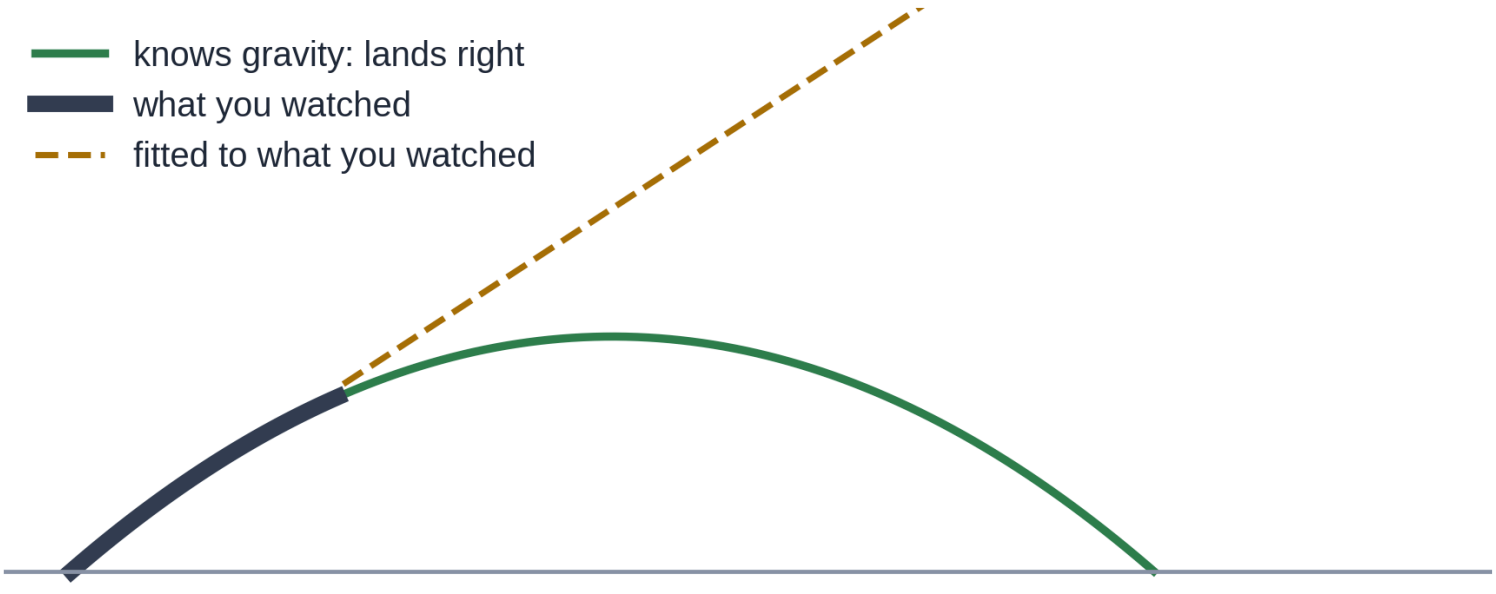


KEY IDEA

Beyond the data: examples only, **27.6 times** worse. Told the rule, **no worse at all** (1.0). The rule was true out there too.

GUESSING FURTHER IS NOT KNOWING FURTHER

Watching half the flight is not enough



EVERYDAY ANALOGY

Copy the shape of what you saw, and the ball keeps rising for ever. Know gravity, and it lands in the right place.

NO FRICTION: THE ENERGY SHOULD NEVER CHANGE

It keeps something nobody asked for

16.3%
drift · examples only

5.1%
drift · examples and the rule

0.2%
drift · the rule only

KEY IDEA

Energy was in nobody's score. Obey the rule, and you inherit everything the rule implies, for free.

ASK A QUESTION THAT WAS NOT ON THE PRACTICE SHEET

Memoriser or understander?

The memoriser

learned the practice answers by heart

The understander

knows the subject

KEY IDEA

Energy was that new question for our machines. Habit: find something the rule implies that you never trained on, and check it.

Part 3, done

★ REMEMBER FOR THE WEEK 6 QUIZ

- Beyond the data: examples only, 27.6 times worse; told the rule, no worse (1.0).
- A machine that only saw examples fails quietly beyond them. It does not know that it does not know.
- Energy was in nobody's score. Examples only drifted 16.3%; the rule only, 0.2%.
- Obey the rule and you inherit everything it implies. Check something you never trained on.

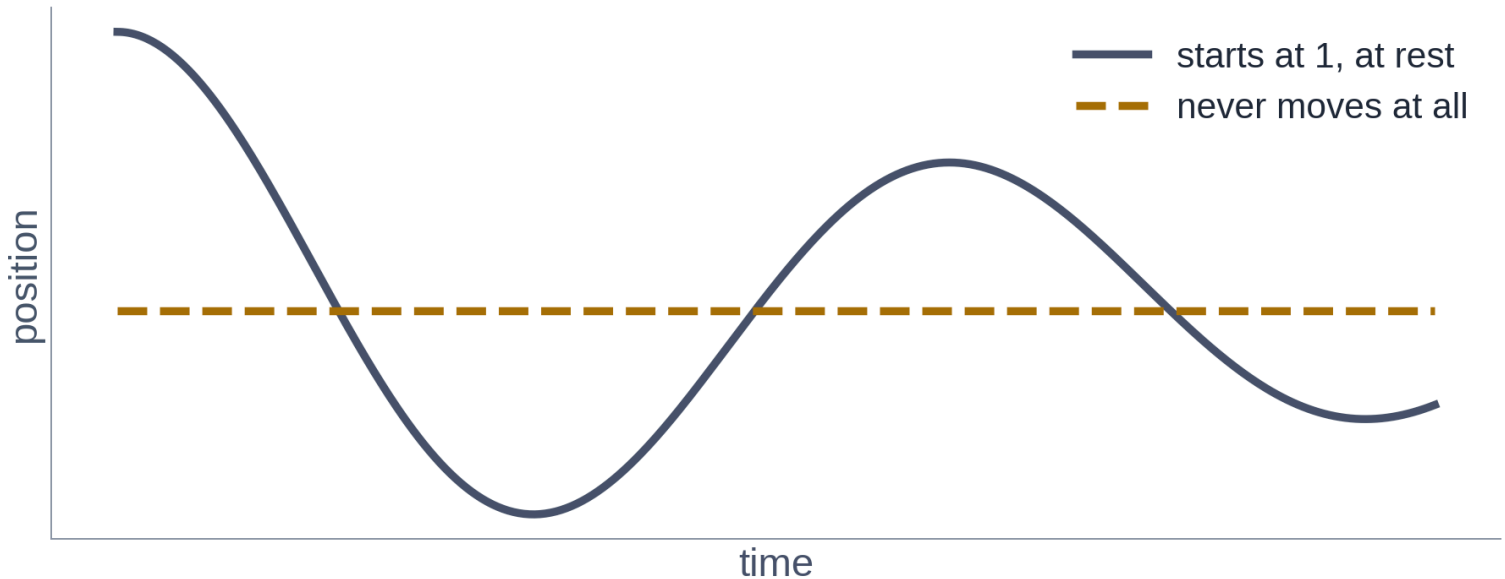
PART 4

What it costs, and how it goes quietly wrong

The most important part of today

WHERE IT STARTS MATTERS

The rule alone does not choose the motion



KEY IDEA

A weight that never moves also obeys the rule: 0 + 0 + 0 = 0. Only the start tells the two lines apart.

ASKING FOR THE START WITH A WEIGHT

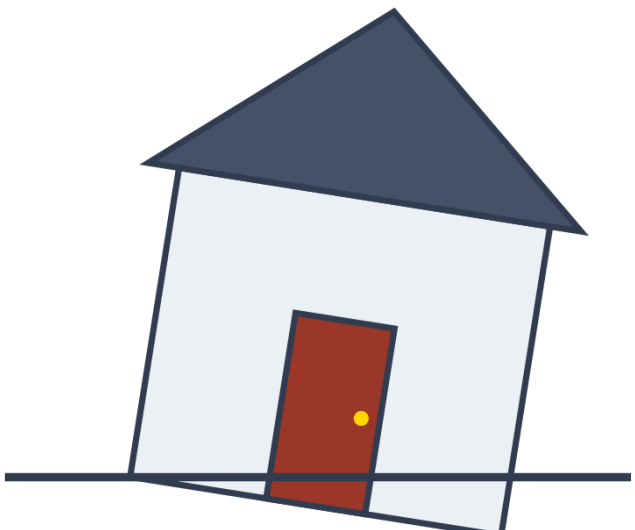
A weight nobody can work out in advance

how the start is given	starts at (should be 1)	error
asked for, weight 1	0.99988	**0.0015**
asked for, weight 50	0.99941	**0.32**
asked for, weight 500	0.99953	**0.66**
built into the sum	1.00000	**0.00135**

KEY IDEA

All four start in the right place. The bad ones got the start perfect and neglected the rule. Nobody can choose the weight in advance.

Badly weighted instructions



perfect door, wobbly house

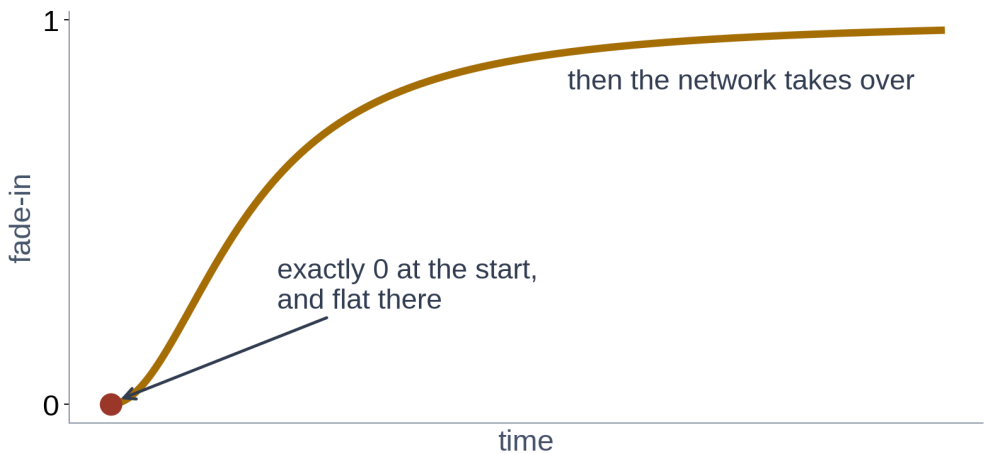
EVERYDAY ANALOGY

Tell a builder the red door is **500 times** more important than the house standing up. You get a perfect door on a leaning house. The builder followed the instructions. The instructions were badly weighted.

MAKE IT IMPOSSIBLE TO GET WRONG

Better: build the start into the sum

> answer = start + fade-in × what the network says



The fade-in is zero at the start, so the start is always exact, whatever the dials do. One wish left; nothing to balance.

The code: the start built in

week5-lab/pinn.py — python

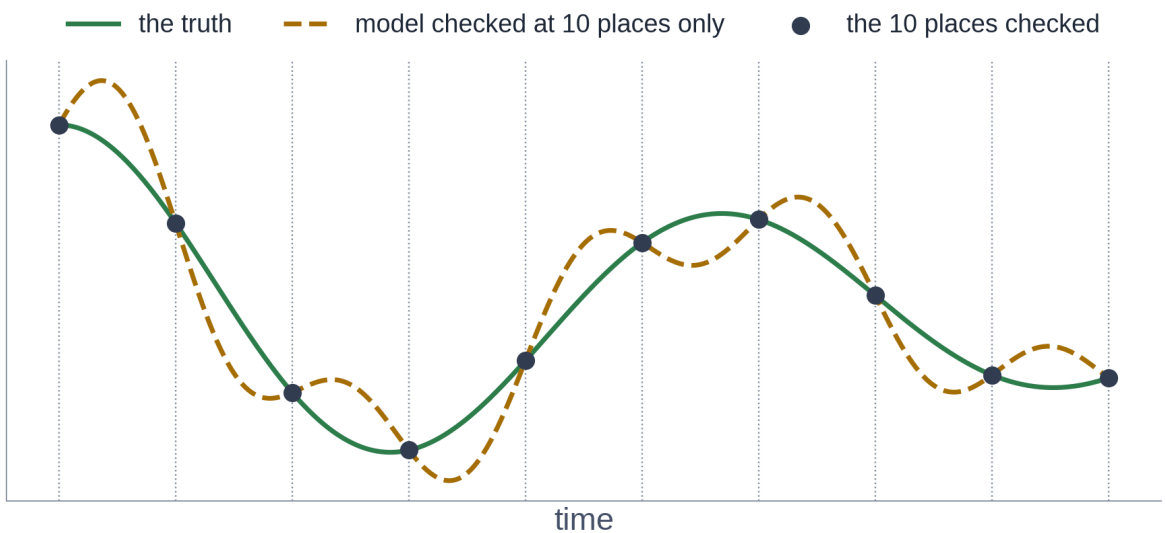
```python
def answer(t, network):
    fade_in = t*t / (1 + t*t)    # 0 at the start, flat there
    return start + fade_in * network(t)
```

KEY IDEA

At the first moment, the network is multiplied by zero, so the answer is exactly the start. Every time.

A SMALL SCORE PROVES LESS THAN YOU THINK

The score only covers the places checked



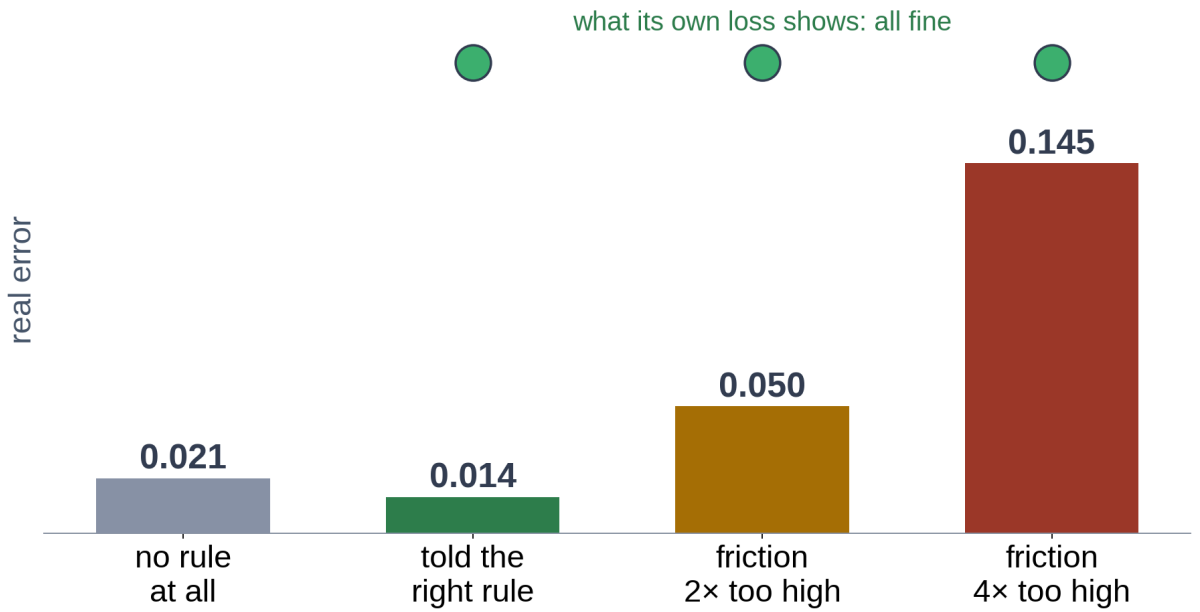
checked at	score	real error
10 places	0.0000026	**0.30**
50 places	0.000049	**0.0015**
200 places	0.000087	**0.0035**

KEY IDEA

The tiniest score came with the biggest error. The score only answers "does it obey the rule where you checked?"

THE MOST IMPORTANT SLIDE OF THE DAY

Wrong rule, and nothing warns you



REAL FRICTION 0.2 · TOLD 0.8

Seven times worse than using **no rule at all**, while its own score stays small. You can only catch it against the real world.

WHY THIS ONE IS SO DANGEROUS

The confident expert

EVERYDAY ANALOGY

An honest beginner says "I'm not sure", so you check. A confident expert with one out-of-date fact gives a precise, well-argued answer, and everything fits together, which is exactly why nobody notices.

A wrong law is worse than no law.

And the score will not tell you.

Part 4, done

★ REMEMBER FOR THE WEEK 6 QUIZ

- The rule alone does not pick the motion; the start does.
- Asking for the start with a weight: errors from 0.0015 to 0.66, and nobody can choose the weight in advance.
- Building the start into the sum: always exact, nothing to balance.
- A small score only speaks for the places checked. A wrong rule is worse than none, and the score stays small.

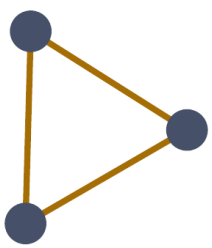
PART 5

Four more ways to build in science

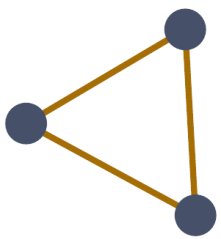
Symmetry, the next-step rule, families, and picture generators

SYMMETRY, THE ONLY GUESS THAT IS EXACTLY TRUE

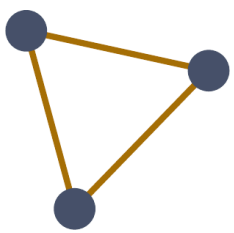
Turn it round: the distances do not change



as measured



rotated 63°



rotated 137°

All three have the same energy.

Raw coordinates:
all six change.

Pairwise distances:
identical to 16 digits.

KEY IDEA

A machine that reads only the distances cannot be fooled by turning: nothing it reads changes when you turn it.

TURN THE PARTICLES ALL THE WAY ROUND

Measured: one wobbles, one cannot

the machine reads	answer changes when turned	worse at angles never seen
positions	**6.4%**	**6.2 times**
distances only	**0.0%**	**1.0 times**

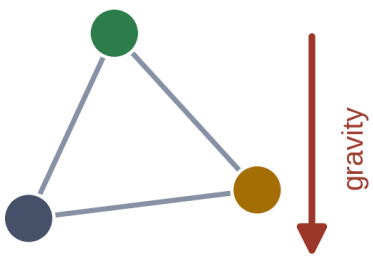
KEY IDEA

The second machine's change is just the computer's rounding: the same number computed twice.

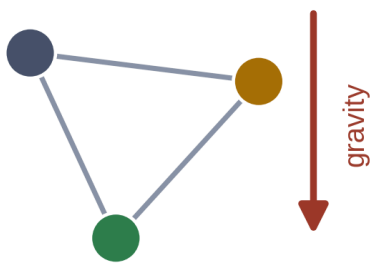
ADD GRAVITY, AND WHICH WAY UP MATTERS

A false symmetry is a wall

this way up



upside down



0.11

reads positions: learns "which way up"

0.24

reads distances: stuck at any amount of data

KEY IDEA

A built-in symmetry is a promise about the world. True: a perfect guarantee. False: a wall the machine can never climb.

The code: read only the distances

week5-lab/symmetry.py — python

```python
def distances(balls):
    return [dist(balls[0], balls[1]),
            dist(balls[0], balls[2]),
            dist(balls[1], balls[2])]

answer = network(distances(balls))   # never sees positions
```

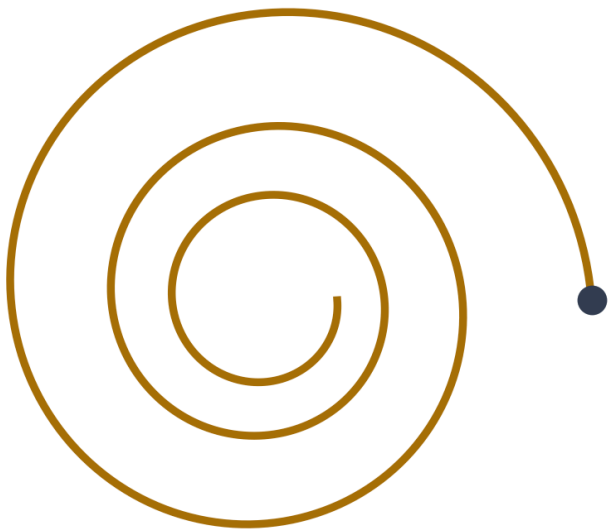
KEY IDEA

The symmetry comes from what the machine is allowed to read, not from training.

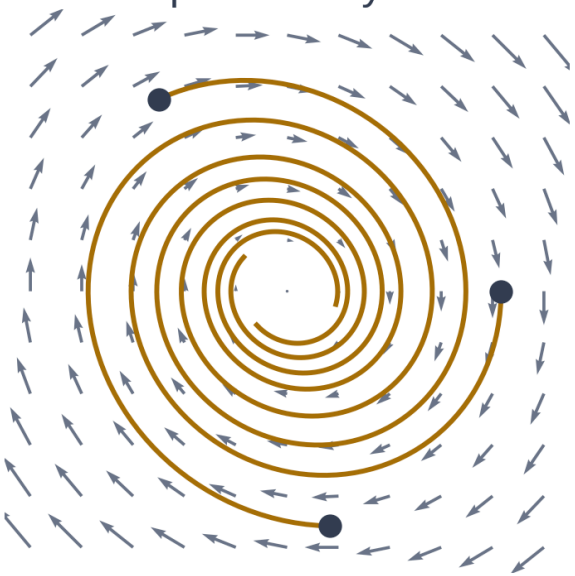
A ROUTE IS ONE JOURNEY; A MAP LETS YOU START ANYWHERE

Learn the rule, not the route

a route: one journey



a map: start anywhere

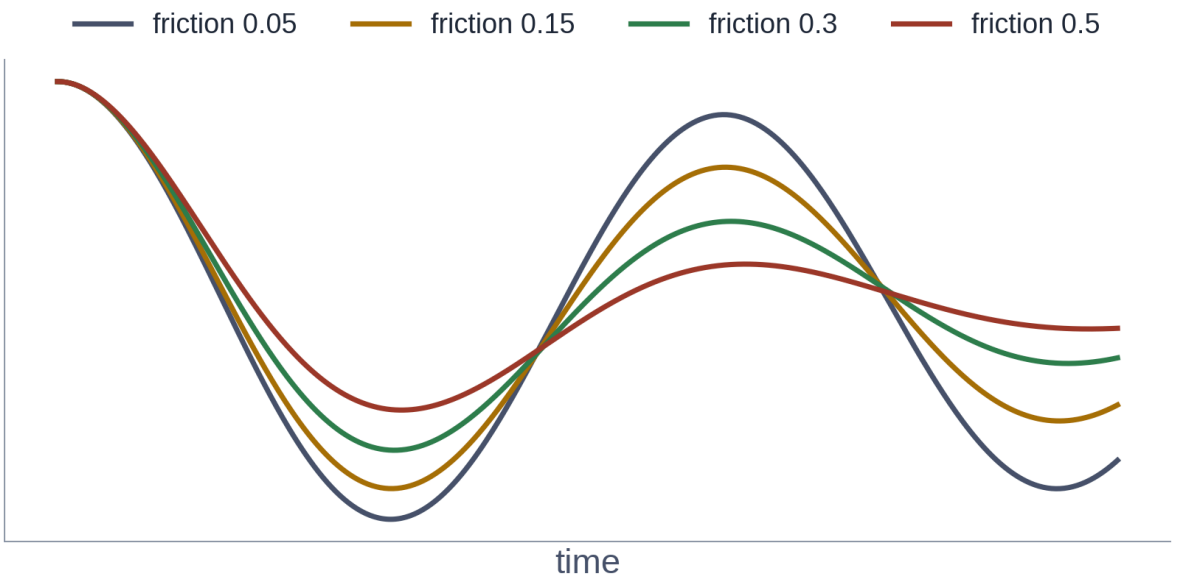


KEY IDEA

Carried three times further than training: learned one motion, **11.6 times** worse. Learned the rule for the next step, **1.7 times**.

THE SAME RULE WITH DIFFERENT NUMBERS IN IT

Solve a whole family at once



KEY IDEA

Trained once on the whole family: a new spring in **a quarter of a millisecond**, instead of seconds from scratch. Faster, rougher: a trade.

A KNOWN PHYSICAL PROCESS, RUN IN REVERSE

Picture makers run ink-in-water backwards



forwards: given by physics, learned by nobody

backwards: one small denoising step, learned, applied many times

KEY IDEA

Physics dissolves the picture into noise. The machine learns only one thing: how to undo one small step. Repeat, and a picture appears.

Part 5, done

★ REMEMBER FOR THE WEEK 6 QUIZ

- Symmetry: read only what the change cannot affect (distances), and the guarantee is exact.
- A false symmetry is a wall: the machine cannot learn the truth at any amount of data.
- Learn the rule for the next step, not one route: then you can start anywhere.
- Learn a whole family at once (fast, rougher). Picture generators undo physics one small step at a time.

PART 6

Choosing

Match the method to what you know

THE BEST RESULT IN THE LAB

Heat along a metal rod, from no data

0.0004

told the rule · shown nothing

0.018

shown 128 measurements

KEY IDEA

Position and time change together, the start and both ends are built into the sum, and the rule alone does about 40 times better.

IF YOU HAVE THIS, REACH FOR THAT

Match the method to what you know

if you have	reach for
a rule, one situation	**a rule-checked network**
a rule, thousands of situations	**learn the whole family**
a rule on a simple shape	**a classical solver**
a rule with an unknown number	**a rule-checked network, run backwards**
half the rule known	**learn the next step, keep the known half**
an exact symmetry	**read only what cannot change**
no reliable rule	**ordinary machine learning**

The claim worth being careful about

Not a better way to solve standard equations.

> **KEY IDEA**
>
> It **is** a better way to: combine a little data with a known rule · find an unknown number · handle awkward shapes · make fast stand-ins for slow calculations.

For a normal equation on a simple shape, sixty years of classical solvers are faster, more accurate, and come with guarantees.

FOR ANYTHING WITH A RULE IN IT

Seven questions before you build

1. What do I know, in plain words?
2. Is it exactly true, roughly true, or a guess?
3. Built into the sum, or asked for with a weight?
4. Where is the score checked, and where not?
5. What does the rule imply that I never trained on?
6. What if my rule is 20% wrong?
7. Would an older method, or ordinary learning, do better?

Part 6, done

★ REMEMBER FOR THE WEEK 6 QUIZ

- Match the method to what you know, not to what is famous.
- A rule on a simple shape: use a classical solver. No reliable rule: ordinary machine learning.
- These methods are better for: a little data plus a rule, unknown numbers, awkward shapes, fast stand-ins.
- Seven questions, starting with: what do I know, and is it exactly true?

ABOUT FIVE AND A HALF HOURS · STOP WHEN THE LIST ENDS

Your week, in order

1	**This lecture.**	⏱ 2 h
2	**Core booklet.** Same six parts, with the pictures.	⏱ 45 min
3	**The lab:** six exercises. Predict, run, explain; write code in two.	⏱ 2 h 35 min
4	**Upload** `SUBMISSION.md`, then compare it with the reference answers.	⏱ 10 min

Labs are not graded. Stuck for 40 minutes? Post in the channel.

THE COMPLETE LIST · IF IT IS NOT HERE, IT WILL NOT BE ASKED

What the Week 6 quiz can ask about Week 5

★1	A pattern nudges; a law rules things out.
★2	Told the rule, shown nothing, it won.
★3	The leftover, the score, and the one changed step.
★4	Beyond the data, and energy for free.
★5	Weights against building the start in; small scores.
★6	A wrong law is worse than none.
★7	Symmetry, the next-step rule, families, picture generators.
★8	Matching the method to what you know.

Put the answer back into the question.

A rule you can check needs no answer sheet.

Questions now, or in the channel any time this week. Next week: the quiz.